Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

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Abstract

Bayesian networks (BNs) can support explainable software engineering (SE) decisions, but constructing their conditional probability tables (CPTs) is costly. Large language models (LLMs) offer a potential source of probability judgments. We investigated whether combining those judgments with the Ranked Nodes Method (RNM) could reduce elicitation effort while retaining agreement with reference CPTs. We compared complete-table elicitation with eliciting selected distributions and reconstructing the remainder through RNM. The sample contained 61 ordinal conditional nodes from seven SE BNs. We also tested structured probability prompts and qualitative retrieval-augmented generation (RAG), producing eight conditions with five replications using GPT-6 Astra at medium effort. To test reconstruction, we compared both methods on configurations omitted from RNM elicitation. This comparison covered 25 nodes: 14 in Scrum, a development-process assessment BN, and 11 in Chrome, two code-review BNs. RNM reduced total variation distance from 0.3184 to 0.2761 for the RNM-derived Scrum reference, but increased it from 0.2223 to 0.2517 for Chrome; the directions persisted across replications. Structured prompting and qualitative RAG produced no consistent improvement across complete tables. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. These findings suggest that compact LLM elicitation can reduce output effort, while its numerical fidelity depends on the reconstruction assumptions. For SE applications, choosing RNM therefore requires checking both those assumptions and agreement with relevant reference probabilities.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a dependable model. Decisions such as release readiness, quality assessment, and process improvement illustrate why both matter. Decision makers need to understand a model's assumptions to judge whether its recommendation applies to their situation. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Yet the observations needed to train machine learning and deep learning models may be scarce, particularly for managerial decisions. Rique et al.'s practitioner study identifies data unavailability as an obstacle to software analytics and notes that collecting the required data can be infeasible [23]. An approach that makes assumptions explicit and incorporates domain knowledge is therefore valuable in these settings.

Bayesian networks (BNs) offer such an approach. A BN represents relationships among variables in a graph and quantifies uncertainty through probabilities, allowing domain knowledge to contribute when historical data are limited [10, 11]. Making these relationships and probabilities explicit gives stakeholders something concrete to inspect and revise. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. This flexibility makes BNs relevant to SE decision support, but building a credible model still requires substantial knowledge-engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. Examples include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their simplifying assumptions reduce elicitation effort, but can also limit how closely the reconstructed table represents the intended probabilities [2].

Large language models (LLMs) are now being explored as a source of the knowledge needed to construct CPTs. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. These developments raise a practical question: should an LLM specify every CPT row, or provide a smaller set of judgments for a semi-automatic method to complete? The latter could reduce the requested output, but any benefit depends on the quality of both the LLM judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

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We also examined whether better support for elicitation could improve either method. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. These treatments tested whether the way a judgment was requested, or the context available to the LLM, improved its numerical agreement. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. Each was repeated five times using GPT-6 Astra at medium effort. We addressed three research questions:

- **RQ1—Reference agreement:** Does RNM reduce total variation distance relative to direct elicitation on the same non-anchor configurations?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?
- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

RQ1 focuses on configurations the LLM was not asked to fill in the RNM condition: these rows test whether reconstruction can replace direct judgments. Twenty-five nodes had at least two parents and at least one such row. All 61 nodes contributed to the complete-table and effort analyses. Our contribution is a controlled comparison of these two construction approaches, including checks that distinguish the LLM's elicited distributions from the tables produced by fitting. RNM improved agreement for the Scrum process-assessment reference but worsened it for the Chrome code-review references. Neither the prompting and evidence treatments nor fewer requested distributions guaranteed better agreement or proportional total-token savings. These results identify conditions to examine when using LLMs to parameterize SE BNs.

2 Related Work

LLM-assisted BN construction varies in what the model supplies, what evidence it receives, and how the resulting probabilities are validated. Table 1 compares the closest CPT-parameterization studies and two adjacent approaches to BN modeling and inference. These distinctions matter because agreement with a reference CPT, usefulness to a modeler, and downstream predictive performance answer different questions.

2.1 Direct elicitation and evidence-based reconstruction

Nafar et al. provide the closest broad benchmark of numerical LLM parameterization [18]. Requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models while requiring more queries. Their FullDist baseline requests one conditional row per query; our direct baseline requests a complete CPT in one call. Their results therefore motivate studying elicitation granularity, but do not resolve whether omitting entire parent configurations is beneficial.

Gottal and Matthes instead reconstruct CPTs from numerical summaries, comparing algorithms with direct LLM output [12]. Their empirical constraints differ from our qualitative RAG packets, which exclude reference probabilities and numerical parameters. In a recent preprint, Rahul and Chowdhury aggregate responses from a virtual persona panel [22]. Their consistency checks include refitting data sampled from the elicited BN itself; these checks cannot establish agreement with an independent reference. The study nevertheless reports computational effort, showing why probability generation should be assessed alongside its resource demands.

2.2 Modeling support and probabilistic reasoning

BIRD illustrates a neurosymbolic approach [7]: LLM judgments are converted into parameters constrained by a BN representation [9]. It is a relevant predecessor for combining elicitation and fitting, although relative support judgments and downstream accuracy do not establish exact CPT recovery. Cypko et al.'s clinical RAG study similarly supports the feasibility of assisting BN modelers without validating complete numerical tables [6]. AutoElicit elicits priors for predictive-model parameters and evaluates prediction after Bayesian updating [4]; its endpoint is distinct from reference-CPT agreement. We evaluate that narrower endpoint with fixed graphs and definitions, without inferring explanation quality or practitioner benefit.

2.3 Partial elicitation and the remaining comparison

RNM constructs ordinal CPTs from weighted parent functions and truncated-normal distributions [11, 15]. InterBeta offers another route to reconstructing tables from partial expert input [1]. Blomaard et al. compare RNM, functional interpolation, and InterBeta using known CPTs to optimize parameters [2]; a recent preprint compares human-elicited WSA and RNM pipelines in software decisions [20]. Our experimental RNM fit instead uses only LLM anchors. Reference-informed fits are separate approximation diagnostics.

These strands leave a specific question: can sparse LLM elicitation retain reference agreement while reducing effort when a constrained method supplies the remaining CPT rows? We examine it through matched direct/RNM comparisons, prompt and qualitative-RAG treatments, and measured token usage. Comparing the elicited anchor distributions with their fitted counterparts shows whether reconstruction improves or worsens agreement on the same configurations; evaluating unqueried rows then assesses the completed table. This design tests an accuracy–effort tradeoff on SE references; it does not assume that an ordinal CPT is well represented by RNM or that fewer requested distributions imply proportional token savings.

3 Method

We conducted a repeated, factorial comparison of direct CPT elicitation and LLM-assisted RNM reconstruction. The graph, variable meanings, state mappings, elicitation templates, and numerical implementation were fixed before formal collection. Our hypotheses predicted lower non-anchor error with RNM (RQ1), improved agreement from structured prompting and qualitative evidence (RQ2), and fewer requested distributions and output tokens with RNM (RQ3).

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; 25 primary targets; reference distances, five replications, tokens, and fitting time.	Compares both pipelines on identical non-anchor rows; separately compares raw and fitted judgments on identical anchor rows.

3.1 Sample and reference construction

We selected published SE BNs for which interpretable node definitions and usable numerical references were available. This availability-based sample covers several applications without representing their prevalence in SE. Table 2 describes the 61 retained conditional targets in seven BNs from five model/source families. Eligibility required a complete numerical reference, interpretable definitions, and an ordinal interpretation for the child and every parent. Binary variables qualified only when their meanings supported an order; nominal families and Boolean gate chains were outside scope. The references cover Scrum practices [19], Chrome code review and defects [14], ERP data quality for advanced planning and scheduling (APS) [13], software maintenance logs [8], and software strategic indicators (SESSI) [16]. The two Chrome versions share one publication family, as do the two software-log BNs.

The Scrum reference implements the published framework for assessing Scrum-based development methods [19]. It relates team and process factors to sprint and backlog quality, allowing project-specific practices and metrics to complement the base model. We evaluated the archived XML version with 25 conditional nodes and 17 roots, using three ordinal states; the published framework describes 44 nodes and five states. Node meanings were operationalized as contextual English descriptions. The evaluated numerical parameters come from the XML, which stores aggregation functions, weights, variances, and expressions. We reconstructed its 249 conditional rows using equal-width intervals and five endpoint-inclusive integration points per parent state, preserving two piecewise expressions and the storage order of a volatility variable. Thus, Scrum provides an RNM-derived reference, whereas the other models provide stored CPTs.

Table 2: Retained targets and requested rows per prompt/evidence combination and replication. Primary targets have at least two parents and at least one non-anchor row.

Network	Targets	Primary	CPT rows	Anchors
Scrum	25	14	249	123
Chrome 40	15	6	78	57
Chrome 42	12	5	63	46
APS	3	0	8	8
SESSI	1	0	3	3
Software logs 3	4	0	16	12
Software logs 4	1	0	4	3
Total	61	25	421	252

For the other BNs, complete exported CPTs define the reference; their publications supply node meanings and semantic ordering. The SESSI target uses the original elicited table in Appendix A3 [16]. References were checked for dimensions, normalization, and parent/child indexing. The largest retained CPT has 27 rows.

The primary comparison tested whether RNM could reconstruct distributions for parent-state combinations omitted from its prompt. We therefore required at least two parents and at least one unqueried configuration, yielding 14 Scrum and 11 Chrome targets. Direct elicitation provided predictions for those same configurations. The other 36 targets remained in the complete-table and effort analyses: some had only one parent, while others had no unqueried rows. This separates the test of multi-parent reconstruction from the broader assessment of the resulting CPTs. References describe particular modeling contexts; they are not universal probabilities for all companies.

3.2 Elicitation conditions and qualitative evidence

The $2 \times 2 \times 2$ design crossed method (direct/RNM), prompt (baseline/structured), and evidence (no RAG/qualitative RAG). These factors distinguish how many distributions the LLM supplied, how the task was framed, and what domain evidence it received. All conditions used the same expert-role description, setting, child and parent definitions, state labels, stored state order, and ordinal ordering. They also specified the assumed direction of each parent relationship: whether a higher parent state should tend to increase or decrease the child state. The requested distribution conditions on *all* listed parent states and averages over unspecified circumstances. It neither fixes omitted circumstances to a middle state nor assumes independence. A persona supplies a professional perspective, without asserting access to company observations.

The baseline asked for normalized distributions in a fixed JSON schema. The structured bundle additionally made conditioning explicit, framed probabilities as expected counts among 100 hypothetical comparable cases (fractional counts allowed), distinguished evidence from assumptions, and requested consistency and normalization checks. These cases are a judgment aid, not observations. We requested a brief evidence note, without collecting private reasoning traces. Because the instructions form one treatment, their individual effects cannot be separated.

Direct elicitation requested a probability vector for every CPT row in one initial call. RNM requested vectors only for predetermined anchors and asked the LLM to choose how parent states should combine: a weighted mean, weighted minimum, weighted maximum, or minimum–maximum mixture. These rules represent different degrees of compensation between stronger and weaker parents. The LLM therefore supplied the anchor probabilities and the rule, while a separate numerical procedure fitted weights and variance to those probabilities (Section 3.3). For a node with only one parent, no combination rule was needed: the parent value itself became the distribution mean, with its direction aligned to the child. Both methods received the same assumed directions, but only RNM enforced them mathematically. Of 92 parent relationships, nine directions followed from variable definitions, 43 were supported by qualitative context, and 40 were explicitly labeled uncertain hypotheses before collection. Errors in these assumptions could therefore affect the reconstructed tables.

RAG supplied short, fixed evidence cards: curated paraphrases of domain explanations associated with the target and its parents. For example, the following complete card accompanied the Scrum Cohesion node:

> **Shared goals.** Cohesive teamwork includes commitment to shared tasks and team goals. Agile collaboration involves joint responsibility, rather than treating every task as an isolated individual obligation.

This card explains the meaning of cohesion and collaboration; it provides no frequency for Low, Medium, or High cohesion. It illustrates the treatment's purpose: to add relevant qualitative context to probability elicitation.

Before collection, the retriever filtered cards by model context and scored overlap between their variable tags and the target-plus-parent set; general-purpose tags contributed 0.25. It selected up to three positive-scoring cards, breaking ties by identifier. Each node received one to three cards, totaling 24–103 words (median 72) within a 220-word budget, and the same packet was reused across paired conditions and replications. Cards excluded reference probabilities, priors, weights, variances, numerical source results, and links identifying reference artifacts. Both evidence conditions already contained the variable definitions, so the comparison measured the added value of qualitative context. It did not test live web retrieval or company-specific frequencies, and input-token budgets were not equalized.

3.3 Ranked-node reconstruction

Following the ranked-node construction [11, 15], each variable with c states partitions $[0, 1]$ into c equal intervals. Negative parent influences reverse semantic rank before aggregation; fitted probabilities are mapped back to the stored child order. Anchors include all-low, all-high, all-middle (rank $\lfloor (c-1)/2 \rfloor$ for each parent), each one-high/rest-low configuration, and each one-low/rest-high configuration, with duplicates removed. Selection uses no reference probabilities. For example, two ternary parents give nine rows, of which five are anchors: $(0,0)$, $(2,2)$, $(1,1)$, $(2,0)$, and $(0,2)$. RNM reconstructs the other four; direct predictions are evaluated on those same four for RQ1.

For k parents, let G_r be the Cartesian product of five equally spaced, endpoint-inclusive points within each parent interval for row r. With standard-normal CDF Φ, child interval $[a_s, b_s]$, aggregation mean μ, and variance v, define

$$T_s(\mu, v) = \frac{\Phi((b_s - \mu)/\sqrt{v}) - \Phi((a_s - \mu)/\sqrt{v})}{\Phi((1-\mu)/\sqrt{v}) - \Phi(-\mu/\sqrt{v})}. \tag{1}$$

The predicted probability is $q_{rs} = 5^{-k} \sum_{x \in G_r} T_s(f(x; w), v)$. Each truncated distribution is normalized before averaging. The mean functions are

$$f_{\text{mean}}(x; w) = \frac{\sum_i w_i x_i}{\sum_i w_i}, \tag{2}$$

$$g_i(x; w) = \frac{w_i x_i + \sum_{j \neq i} x_j}{w_i + k - 1}, \tag{3}$$

$$f_{\min} = \min_i g_i, \qquad f_{\max} = \max_i g_i, \tag{4}$$

$$f_{\text{mix}}(x; a) = (1-a) \min_i x_i + a \max_i x_i. \tag{5}$$

For a single parent, the mean is its aligned value x_1. Weighted-mean log weight ratios relative to the last parent lie in $[-6, 6]$;

weighted-minimum/maximum weights lie in $[1, 20]$; $a \in [0, 1]$; and $v \in [10^{-4}, 1]$. Variance and minimum/maximum weights use logarithmic parameterizations.

For the family selected by the LLM, the fitter minimized mean anchor total variation between fitted and elicited distributions. It used eight deterministic starts (seed 42), coordinate-search steps 2^{-2} through 2^{-11} on normalized parameter coordinates, at most 20 sweeps per step, and an improvement threshold of 10^{-12}. Ties retained the earlier candidate. This seed controls fitting, not LLM sampling. The fitter received no reference CPT input. Poor but valid fits were retained; similar tables can arise from different parameters, so agreement does not establish parameter recovery. Reference-informed fits were computed separately after predictions were fixed, solely to diagnose approximation limits.

3.4 Execution and reference isolation

Five replications were collected on 15–16 October 2026 through Codex CLI 0.162.1, requesting `gpt-6-astra` with `medium` reasoning effort. Each target–condition–replication used a fresh session. Requests were shuffled within each replication; sampling temperature and generation seed were unspecified. The replication package records the exact request order and configuration.

The experimental computer and sessions had no access to reference files, full papers, evaluator code, or authoring conversations. Browsing, connectors, memory, and tools were disabled. Each request received only its designated definitions and qualitative evidence. Reference probabilities were also excluded from RNM fitting and family selection.

Validation required the correct rows and states, finite probabilities in $[0, 1]$, and row sums within 10^{-6} of one. A failed schema check allowed one fresh-session format repair using the original task, its answer, and the validation errors, without reference feedback. The 2,440 target–condition–replication trials required four repairs, two per method. All produced valid outputs, so no fallback distributions were used; token accounting includes all 2,444 attempts.

3.5 Outcomes and statistical analysis

For reference row p and prediction q, the primary distance is

$$\mathrm{TV}(p, q) = \tfrac{1}{2} \sum_s |p_s - q_s| \in [0, 1], \tag{6}$$

where lower values indicate closer reference agreement. RQ1 compared identical non-anchor configurations for both methods. Within each replication, we averaged rows within targets, targets within BNs, and BNs within publication families, then averaged the four prompt and evidence combinations equally. Scrum is an equal-target mean reported separately; Chrome gives its two versions equal weight. We do not pool their primary effects.

RQ2 used all rows and contrasted structured minus baseline or RAG minus no RAG within each method, averaging the other factor equally. Scrum remains separate; the external SE macro-average gives equal weight to four publication families. It is broader than the Chrome-only primary comparison. Secondary calculations retain Jensen–Shannon divergence in bits, mean absolute ordinal-CDF differences, and smoothed row-weighted CPT KL divergence. KL adds 10^{-8} to each state and renormalizes; its row weighting differs from the primary macro-average and it is not joint-distribution KL. TV is the primary endpoint.

For RQ3, we counted requested vectors and scalars before repairs, and aggregated client-reported input and output tokens across *all attempts*, including repairs. Total tokens are input plus output; cached-input and reasoning-output counters are subsets and are not added again. Paired effort contrasts use each method's replication totals, both for all 61 targets and the 25 primary targets. Summed request latency, repairs, and recorded offline fitting durations provide additional effort descriptions. Token counts from subscription execution are not converted into API-priced monetary savings.

Each replication repeated every condition on the same targets, providing a paired method comparison. We used these five replication blocks as the inferential units, rather than treating nodes or rows from shared BNs as independent observations. We used two-sided paired Student t-tests with four degrees of freedom and pointwise 95% t intervals on block differences. These concern repeated executions on the fixed benchmark, assuming approximately normal, independent, stationary differences. Fresh sessions and shuffled order do not establish those assumptions, and five blocks provide little distributional evidence. A two-sided exact sign test checks sensitivity to block directions under independence; it addresses a direction/median null rather than exactly the same mean null. Even five concordant nonzero differences yield $p = 0.0625$.

Holm adjustments were applied separately to two primary contrasts, eight prompt/RAG contrasts, four token contrasts, and two same-anchor fitting contrasts. This is exploratory family-wise adjustment, not study-wide confirmatory control; intervals are not simultaneous. Other family and sensitivity comparisons are descriptive. No smallest effect of interest or equivalence margin was fixed, so significance does not establish practical usefulness and nonsignificance does not establish equivalence. No inference to arbitrary companies or unseen BNs is warranted.

3.6 Post-hoc offline diagnostics

To examine whether the observed contrast arose from the elicited judgments, the chosen combination rule, or the numerical search, we conducted additional analyses using the saved responses. We first compared both methods with uniform probabilities and checked secondary distance measures, then varied the reconstruction while preserving the original anchors and predictions. One variant fitted direct judgments on paired RNM anchors with the RNM-selected family; others selected among four families by LLM-anchor TV, increased search effort, or expanded bounds. The supplement specifies search settings and bounds. Directions and five-point integration remained fixed; no reference probability entered fitting or selection.

Separately, reference-informed diagnostics minimized complete-reference TV across four families under stronger search, expanded bounds, and every combination of reversed parent directions. These in-sample best-found approximations assess compatibility with the reference; they are not held-out predictions or certified error floors.

We also report Chrome version means and leave-one-node-out diagnostics, re-averaging remaining targets within BNs with equal Chrome-version weights.

4 Results

The results reveal a tradeoff: RNM asks the LLM for fewer judgments, but whether reconstruction improves the CPT depends on the reference model. We first examine agreement on rows that RNM must reconstruct, then ask whether prompting and qualitative evidence change that pattern, and finally compare the computational effort. Lower TV always means closer reference agreement; a negative RNM-minus-direct difference favors RNM.

4.1 RQ1: When does reconstruction improve agreement?

The same method helps Scrum and hurts Chrome. On the 14 primary Scrum targets, mean non-anchor TV decreases from 0.3184 with direct elicitation to 0.2761 with RNM: a reduction of 0.0423, or 13.3%. RNM is closer on 12 of the 14 targets and in each of the four matched prompt/evidence conditions. On the 11 Chrome targets, the result reverses: TV increases from 0.2223 to 0.2517, a rise of 0.0294, or 13.2%. RNM is closer on only three targets. Thus, the hypothesized benefit holds for the RNM-derived Scrum reference, but not for the independently stored Chrome CPTs.

Figure 1a shows that this contrast is present in every replication, rather than being caused by one unusual run. The paired-t intervals for RNM minus direct are $[-0.0486, -0.0360]$ for Scrum and $[+0.0259, +0.0329]$ for Chrome; both Holm-adjusted p-values are below 0.001. However, the exact sign tests do not reject after adjustment ($p = 0.125$ in each stratum). The direction is consistent across these runs, while statistical rejection depends on the paired-t assumptions. These tests do not determine whether the differences are useful for a particular SE decision.

The anchor rows locate a problem in reconstruction. RNM receives explicit LLM distributions for its anchor configurations before fitting a compact function. Comparing those distributions with the fitted probabilities on the *same rows* helps identify what fitting changes (Figure 1b). For Scrum, raw-anchor TV is 0.2790 and fitted-anchor TV is 0.2762: fitting changes agreement little. For Chrome, it rises from 0.2491 to 0.2994. Direct judgments on these Chrome anchors have TV 0.2507, close to the raw RNM value. On the elicited Chrome configurations, fitting introduces additional disagreement beyond that present in the raw judgments.

The contrast is particularly pronounced in Chrome 42. Its primary TV rises from 0.2377 to 0.2933, whereas Chrome 40 changes from 0.2069 to 0.2102. Removing any one target preserves the corresponding Scrum or Chrome direction in all five replications. The next examples show what these changes can look like in a CPT; they illustrate mechanisms rather than typical cases.

Two configurations make the tradeoff concrete. Consider a Scrum team with low self-organization but medium collaboration. The model's Cohesion node asks whether the team works synchronously toward shared goals. For this parent configuration, the reference assigns 76.6% probability to Low cohesion. Direct elicitation assigns 58.0%, while RNM reconstructs 67.6%, moving closer to the reference (Figure 2a). Mean row TV falls from 0.1862 to 0.0901. This is a configuration that the RNM prompt did not ask the LLM to fill: the fitted relationship supplies it from the anchors.

Now consider a Chrome component with three or four contributors and no minor contributors. Contributors are classified as major or minor according to their share of changes. With no minor contributors, all must be major, so the Major Contributors node belongs entirely in the three-to-four category. Both the reference and all direct responses assign probability one to that category. The RNM reconstruction assigns 84.4%, with the remaining 15.6% placed in neighboring categories (Figure 2b). Here, smoothing a table through the fitted family weakens a count constraint that direct elicitation preserves. This target has the largest Chrome disadvantage, but removing it still leaves Chrome's aggregate difference positive at +0.0168.

Alternative fits retain the contrast. Could the result mainly reflect different LLM answers or an insufficient parameter search? Fitting direct judgments on the paired RNM anchors still improves Scrum relative to the original direct pipeline (0.3184 to 0.2924) and worsens Chrome (0.2223 to 0.2456). Selecting the aggregation family by its fit to LLM anchors narrows Chrome's disadvantage to 0.2413, but does not reverse it. Stronger searches and wider parameter bounds make little difference. These post-hoc checks support a mismatch in the tested reconstruction pipeline rather than a simple shortage of optimization effort; they do not isolate every component.

Influence assumptions also matter. Four Chrome targets exhibit mean-direction reversals and six exhibit stochastic-order reversals under the specified parent alignment. Scrum has two targets with reversals. Even fitting directly to complete reference tables yields a higher best-found error for Chrome than Scrum (0.1592 versus 0.0560); allowing wider bounds and reversed directions lowers these to 0.1322 and 0.0260. This reference-informed diagnostic supports a compatibility explanation, without establishing a global optimum or a lower bound on RNM error.

The broader complete-table results cover all 61 targets. TV changes from 0.2605 to 0.2334 for Scrum and from 0.1770 to 0.2153 for the four-family external SE average. Among the latter, Chrome, APS, and software logs worsen, while SESSI improves slightly. The supplement provides the eight-condition means, family results, uniform baseline, secondary metrics, and all sensitivity results. **RQ1 is therefore supported for Scrum and opposed for Chrome: ordinal eligibility alone does not guarantee that RNM reconstruction improves reference agreement.**

4.2 RQ2: Do richer prompts or qualitative evidence help?

The method contrast remains across the prompt/evidence combinations. Table 3 isolates the incremental effect of structured prompting and RAG on complete-table TV. For direct Scrum elicitation, structured prompting lowers TV by 0.0069, while RAG raises it by 0.0043; these are the only two comparisons surviving the eight-test Holm correction. Neither RNM Scrum contrast nor any external SE contrast survives that correction, and none rejects under the exact sign test.

The evidence packets supplied explanations of relationships, not observed frequencies. For example, the Cohesion packet describes shared goals, joint responsibility, and team adaptation; it does not say how often cohesion is Low when self-organization is Low. Both evidence conditions already receive node definitions.

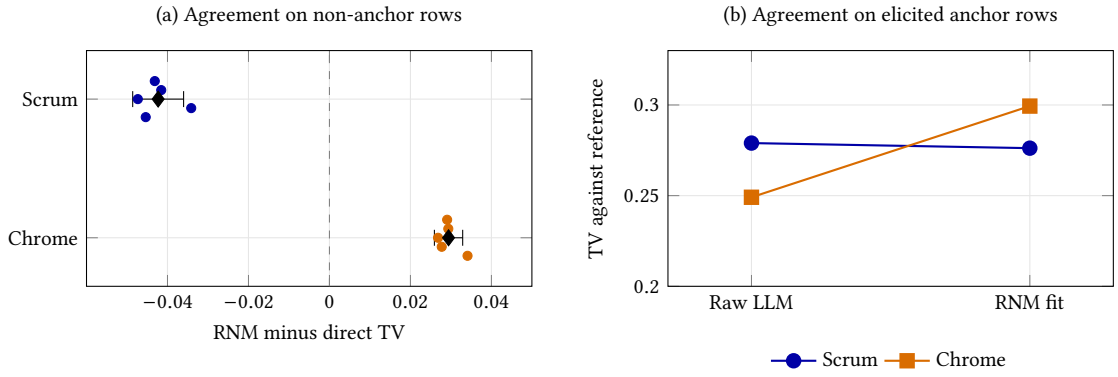


Figure 1: Where the methods diverge. (a) Each small circle is one replication's primary RNM-minus-direct TV difference, averaged across prompt/evidence conditions for 14 Scrum or 11 Chrome targets. Diamonds and whiskers show means and pointwise 95% t intervals; horizontal position carries the result, and vertical jitter separates the five replications. Negative values favor RNM. (b) Mean errors before and after fitting, evaluated on the same anchor rows. Chrome agreement deteriorates during fitting even for configurations supplied to RNM. Panels use different row sets; their errors are not an additive decomposition.

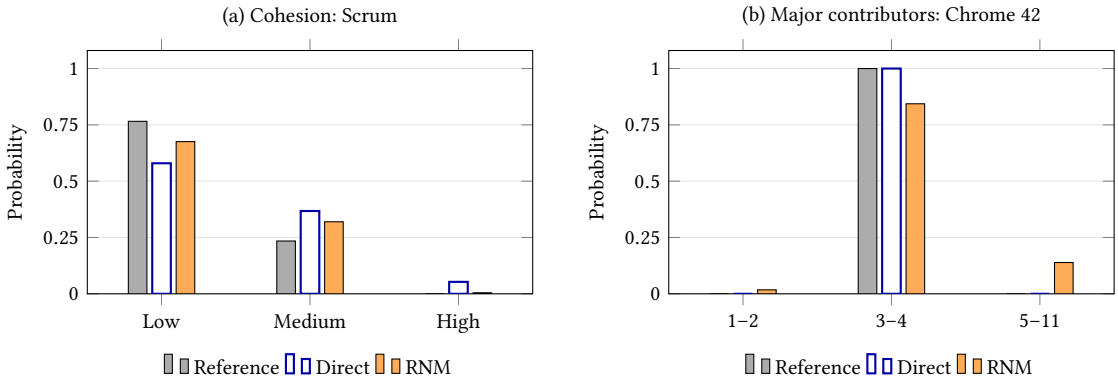


Figure 2: Two worked CPT rows, both omitted from RNM elicitation. (a) Cohesion given Low Self-Organizing and Medium Collaboration. RNM moves probability toward Low, closer to the reference. (b) Major contributors given three or four total contributors and zero minor contributors. The count relationship requires all contributors to be major; direct elicitation preserves this, whereas RNM spreads probability to neighboring categories. Bars average 20 predictions per method. These deliberately contrasting illustrations use the largest target-level improvement and deterioration and the first stored non-anchor row of each; they are not representative cases.

Table 3: RQ2: complete-table TV change from adding structured prompting or RAG, averaging the other factor equally. Negative values favor the addition. Asterisks mark Holm-adjusted paired-t $p < 0.05$ across eight contrasts; no exact sign test rejects. Full intervals and p-values are in the supplement.

Reference	Method	Prompt Δ	RAG Δ
Scrum	Direct	−0.00692*	+0.00430*
Scrum	RNM	−0.00313	+0.00132
External SE	Direct	−0.00462	+0.00163
External SE	RNM	−0.00060	+0.00085

RAG therefore adds qualitative guidance to an already contextualized task, rather than supplying missing numerical constraints. **RQ2 receives no consistent support for the tested prompt bundle and qualitative packets.** This result does not establish that company measurements or quantitative retrieval would be ineffective.

4.3 RQ3: Do fewer judgments translate into fewer tokens?

RNM requests 5,040 distributions across the experiment, compared with 8,420 for direct elicitation: 40.1% fewer. Yet both methods still make one initial request per target and condition. Fewer table entries therefore shorten what must be returned, without eliminating

the shared instructions, definitions, or evidence sent with each request.

This distinction explains why the savings shrink as accounting broadens. Output tokens decrease by 12.12%, but input-plus-output tokens decrease by only 0.61% (Table 4). Inputs dominate the total: approximately 14.36 million for direct elicitation and 14.31 million for RNM. The 25 primary targets, where multi-parent compression is possible, show larger savings: 28.28% in output tokens and 2.94% in total tokens. Even there, the reduction in requested probabilities is not a proportional reduction in input-plus-output tokens.

Output savings survive the paired-t adjustment for both workloads, and total-token savings survive it for the primary targets only. Summed request latency is 3.82% lower for RNM; offline reconstruction adds 182.3 seconds of recorded fit time. These are computational descriptions, not controlled wall-clock or monetary savings. **RQ3 is supported for reduced elicitation and output size, but offers limited evidence of overall token savings.** The observed token savings accompany improved Scrum agreement but worse Chrome agreement: fewer requested judgments do not ensure a better CPT.

5 Discussion

The results connect two questions that are often considered separately: whether an LLM can supply useful probability judgments and whether a compact construction method preserves them. The contrast between Scrum and Chrome shows why the complete construction process matters. We discuss that contrast against prior work, then draw implications for developing and applying SE BNs.

5.1 Research questions in relation to the literature

RQ1: Reference agreement. For RQ1, RNM's improvement on Scrum is consistent with its ability to represent a reference already constructed using ranked nodes [11]; the improvement also persisted when fitting used the direct method's distributions at the same anchor configurations. Chrome's raw anchor judgments are similarly accurate across methods, but fitting worsens agreement. The disadvantage concentrates in Chrome 42; no single target determines its direction. Choosing the combination rule by numerical fit to the LLM judgments narrowed this disadvantage, but a more thorough search and wider parameter bounds did not remove it. This supports the distinction between ordinal eligibility and RNM compatibility [15]. Reference-informed direction changes reduce approximation error, but cannot separate remaining representation and optimization limitations. Chrome's documented interdependencies [14] provide context, not proof of the mechanism. Nor does CPT agreement establish defect-prediction accuracy.

RQ2: Elicitation support. Structured prompting benefits direct Scrum elicitation but not the other adjusted comparisons. Nafar et al. likewise distinguish query-granularity gains from prompt modifications [18]; they changed how individual row probabilities were requested, whereas we reduced the number of rows requested. Their results therefore cannot establish the benefit of RNM reconstruction. Qualitative packets give no consistent benefit beyond shared definitions. Clinical retrieval-usability findings [6] and reconstruction from statistical summaries [12] address different evidence and outcomes. Our results do not establish that informative quantitative retrieval is ineffective.

RQ3: Accuracy and effort. Fewer distributions yield lower output use, but small total-token savings. Extending the human burden–fidelity perspective [1], computational effort includes inputs, reasoning, repairs, and fitting. Compression improves reference agreement on Scrum while worsening it on Chrome, preventing a universal efficiency–accuracy recommendation.

5.2 Implications for SE researchers and practitioners

For researchers, the contrast motivates independently constructed references, validated influence directions, and multiple model configurations. Comparing raw and fitted distributions on the same anchors shows what fitting changes; evaluating unqueried rows tests whether those changes extend usefully to the rest of the table. Hybrid inference such as BIRD [9] also motivates testing structured components rather than inferring fidelity from inspectability. Repeated runs do not establish generalization to other companies.

For practitioners, RNM is a candidate initialization method requiring local validation. Review node meanings and influence assumptions, compare direct and reconstructed judgments, and validate additional configurations with company measurements or experts. A small fitting error only means that the reconstructed table matches the elicited judgments; it does not establish that those judgments describe the organization. Tailoring reusable models remains plausible but requires held-out organizational validation before decision use.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. Separate strata and anchor diagnostics expose this distinction. SESSI represents an original elicitation stage. Adapted variable meanings and uncertain parent directions may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, reference isolation, and blinded fitting limit direct access leakage. Public benchmark memorization remains possible. The logs identify the requested model configuration, but cannot independently verify the exact serving-model version. Prompt and fitting development used the Scrum setting, which may favor that reference. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. The elicitation instructions and evidence cards were tested as bundles, so the contribution of each individual instruction or passage is unknown.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. The SE subset and statistical

Table 4: RQ3: observed tokens over all five replications, including format repairs. Workloads combine reference strata for accounting only. Holm-adjusted paired-t tests compare five method-paired replication totals across four contrasts; no exact sign test rejects. Complete intervals are in the supplement.

Targets	Counter	Direct	RNM	Reduction	Holm t p
All 61	Output	341,153	299,798	12.12%	0.000110
All 61	Input + output	14,697,090	14,607,517	0.61%	0.080921
Primary 25	Output	213,972	153,463	28.28%	0.000019
Primary 25	Input + output	6,384,519	6,196,992	2.94%	0.016292

comparisons were selected after inspecting the broader results; inference is therefore exploratory. Separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The primary comparison contains one external publication family (Chrome) and one Scrum model. The 61-target sample spans more settings, but many targets do not test multi-parent compression, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different trade-offs.

7 Conclusion

This study examined whether LLMs can reduce the effort of constructing SE CPTs by supplying selected probability judgments to RNM instead of filling every row directly. Across 61 ordinal conditional nodes from seven BNs, we compared both approaches under baseline and structured prompts, with and without qualitative RAG, over five replications. The central finding is that fewer elicited judgments do not necessarily produce a better table: the reconstruction method is part of the source of error and must be evaluated alongside the LLM.

The 25-node comparison on configurations omitted from RNM elicitation showed contrasting outcomes. RNM reduced TV by 13.3% for Scrum's RNM-derived reference and increased it by 13.2% for the Chrome code-review references. On Chrome configurations that were elicited, raw LLM judgments had similar agreement across methods, but RNM fitting increased disagreement. The worked contributor-count example illustrates how a fitted distribution can lose a constraint preserved by direct elicitation. Alternative fits and single-node omissions retained the aggregate directions, supporting a limitation of the tested reconstruction approach rather than an explanation based on one exceptional node. Structured prompting and qualitative RAG provided no consistent improvement across the sample.

The efficiency benefit was also narrower than the reduction in judgments suggests. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but input tokens dominated the workload, leaving a 0.61% reduction in total tokens. For SE researchers, this means evaluating elicitation, reconstruction, and resource use together. For practitioners, LLM-generated CPTs are candidates for model initialization that still require validation against the intended organizational context; ordinal state labels alone do not establish that RNM is an appropriate representation.

These conclusions concern agreement with the studied reference CPTs, using one LLM configuration, rather than validated decisions across organizations. The contrasting directions were stable across replications, although statistical rejection depended on the paired-t assumptions. The next steps are to test whether domain-validated influence relationships and organizational measurements improve reconstruction, and whether the resulting BNs support better predictions or decisions on held-out projects. This would connect the numerical comparisons reported here to the practical purpose of SE decision support.

Replication Materials

The replication package contains the elicitation materials, reference CPTs, saved responses, fitted tables, evaluation scripts, token counts, and supplementary analyses. Reference materials are separated from collection inputs. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

References

[1] Martine J. Barons, Steven Mascaro, and Anca M. Hanea. 2022. Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks. *Risk Analysis* 42, 6 (2022), 1196–1234. doi:10.1111/risa.13772

[2] Bodille P. M. Blomaard, Gabriela F. Nane, and Anca M. Hanea. 2025. Approaches for Reducing Expert Burden in Bayesian Network Parameterization. *Entropy* 27, 6, Article 579 (2025), 27 pages. doi:10.3390/e27060579

[3] Sicong Cao, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang, Bin Li, Wei Liu, Di Wu, and Yixin Chen. 2025. A Systematic Literature Review on Explainability for ML/DL-based Software Engineering. *Comput. Surveys* 58, 4, Article 95 (2025), 34 pages. doi:10.1145/3763230

[4] Alexander Capstick, Rahul Krishnan, and Payam Barnaghi. 2025. AutoElicit: Using Large Language Models for Expert Prior Elicitation in Predictive Modelling. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 6746–6777. https://proceedings.mlr.press/v267/capstick25a.html

[5] Kwai-Sang Chin, Da-Wei Tang, Jian-Bo Yang, Shui Yee Wong, and Hongwei Wang. 2009. Assessing new product development project risk by Bayesian network with a systematic probability generation methodology. *Expert Systems with Applications* 36, 6 (2009), 9879–9890. doi:10.1016/j.eswa.2009.02.019

[6] Mario A. Cypko, Muhammad Agus Salim, Aditya Kumar, Leonard Berliner, Andreas Dietz, Matthaeus Stoehr, and Oliver Amft. 2026. Large language models

with retrieval-augmented generation enhance expert modelling of Bayesian network for clinical decision support. *International Journal of Computer Assisted Radiology and Surgery* 21, 2 (2026), 211–222. doi:10.1007/s11548-025-03524-9

[7] Artur d'Avila Garcez and Luis C. Lamb. 2023. Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review* 56, 11 (2023), 12387–12406. doi:10.1007/s10462-023-10448-w

[8] Santiago del Rey, Silverio Martínez-Fernández, and Antonio Salmerón. 2023. Bayesian Network analysis of software logs for data-driven software maintenance. *IET Software* 17, 3 (2023), 268–286. doi:10.1049/sfw2.12121

[9] Yu Feng, Ben Zhou, Weidong Lin, and Dan Roth. 2025. BIRD: A Trustworthy Bayesian Inference Framework for Large Language Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=fAAaT826Vv

[10] Norman Fenton, Martin Neil, William Marsh, Peter Hearty, Łukasz Radliński, and Paul Krause. 2008. On the effectiveness of early life cycle defect prediction with Bayesian Nets. *Empirical Software Engineering* 13 (2008), 499–537. doi:10.1007/s10664-008-9072-x

[11] Norman E. Fenton, Martin Neil, and Jose Galan Caballero. 2007. Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks. *IEEE Transactions on Knowledge and Data Engineering* 19, 10 (2007), 1420–1432. doi:10.1109/TKDE.2007.1073

[12] Jonas Gottal and Florian Matthes. 2026. Verifiable Parameterization of Bayesian Networks from Scientific Literature: Unlocking Unstructured Empirical Evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 3400–3414. doi:10.18653/v1/2026.findings-acl.167

[13] Jan-Phillip Herrmann, Sven Tackenberg, Elio Padoano, Jörg Hartlief, Jens Rautenstengel, Christine Loeser, and Jörg Böhme. 2022. An ERP Data Quality Assessment Framework for the Implementation of an APS system using Bayesian Networks. *Procedia Computer Science* 200 (2022), 194–204. doi:10.1016/j.procs.2022.01.218

[14] Andrey Krutauz, Tapajit Dey, Peter C. Rigby, and Audris Mockus. 2020. Do code review measures explain the incidence of post-release defects? Case study replications and Bayesian networks. *Empirical Software Engineering* 25 (2020), 3323–3356. doi:10.1007/s10664-020-09837-4

[15] Pekka Laitila and Kai Virtanen. 2020. On Theoretical Principle and Practical Applicability of Ranked Nodes Method for Constructing Conditional Probability Tables of Bayesian Networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50, 5 (2020), 1943–1955. doi:10.1109/TSMC.2018.2792058

[16] Martí Manzano, Claudia Ayala, Cristina Gómez, Antonin Abherve, Xavier Franch, and Emilia Mendes. 2021. A Method to Estimate Software Strategic Indicators in Software Development: An Industrial Application. *Information and Software Technology* 129 (2021), 106433. doi:10.1016/j.infsof.2020.106433

[17] Emilia Mendes. 2012. Using knowledge elicitation to improve Web effort estimation: Lessons from six industrial case studies. In *Proceedings of the 34th International Conference on Software Engineering*. IEEE, 1112–1121. doi:10.1109/ICSE.2012.6227108

[18] Aliakbar Nafar, Kristen Brent Venable, Zijun Cui, and Parisa Kordjamshidi. 2026. Extracting Probabilistic Knowledge from Large Language Models for Bayesian Network Parameterization. *Transactions on Machine Learning Research* (May 2026). https://openreview.net/forum?id=Fy3Byg3CVo

[19] Mirko Perkusich, Kyller Gorgonio, Hyggo Almeida, and Angelo Perkusich. 2017. A Framework to Build Bayesian Networks to Assess Scrum-based Software Development Methods. In *Proceedings of the 29th International Conference on Software Engineering and Knowledge Engineering*. 7 pages. doi:10.18293/SEKE2017-139

[20] Mirko Perkusich, João Nunes, Emilia Mendes, Emanuel Dantas, Ademar Sousa, Danyllo Albuquerque, Kyller C. Gorgônio, and Angelo Perkusich. 2026. Semi-Automatic Quantification of Bayesian Networks for Software Decision Support: Comparing WSA and RNM in a Software R&D Organization. *arXiv preprint arXiv:2609.31275* (2026), 37 pages. https://arxiv.org/abs/2609.31275

[21] Mirko Perkusich, Gustavo Soares, Hyggo Almeida, and Angelo Perkusich. 2015. A procedure to detect problems of processes in software development projects using Bayesian networks. *Expert Systems with Applications* 42, 1 (2015), 437–450. doi:10.1016/j.eswa.2014.08.015

[22] Kumar Rahul and Shovan Chowdhury. 2026. Human–AI Construction of Bayesian Networks for Operational Decision Support: A Virtual Survey Approach. *arXiv preprint arXiv:2607.14141* (2026). doi:10.48550/arXiv.2607.14141

[23] Thiago Rique, Mirko Perkusich, Emanuel Dantas, Danyllo Albuquerque, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2023. On Adopting Software Analytics for Managerial Decision-Making: A Practitioner's Perspective. *IEEE Access* 11 (2023), 73145–73163. doi:10.1109/ACCESS.2023.3294823

[24] Thiago Rique, Mirko Perkusich, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2025. Constructing the graphical structure of expert-based Bayesian networks in the context of software engineering: A systematic mapping study. *Information and Software Technology* 177 (2025), 107586. doi:10.1016/j.infsof.2024.107586